

Inference via robust optimal transportation: theory and methods

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Optimal transportation is widely-applied in statistics and machine learning. Despite its popularity, inference based on it has some issues. For instance, it is sensitive to outliers and may not be even defined when the underlying model has infinite moments. To cope with these problems, first we consider a robust version of the primal transportation problem and show that it defines the robust Wasserstein distance, $W^{(\lambda)}$, depending on a tuning parameter $\lambda > 0$. Second, we illustrate the link between 1-Wasserstein distance W_1 and $W^{(\lambda)}$ and study its key measure theoretic aspects. Third, we derive some concentration inequalities for $W^{(\lambda)}$. Fourth, we use $W^{(\lambda)}$ to define minimum distance estimators, provide their statistical guarantees and illustrate how to apply the concentration inequalities for a selection of λ . Fifth, we provide the dual form of the robust optimal transportation and apply it to machine learning problems. We review, in a unified perspective, the key aspects of OT and ROBOT, while complementing the extant ROBOT results with our new methodological findings. Numerical exercises provide evidence of the benefits of our novel methods.

Keywords — Concentration inequalities, Minimum distance estimation, Robust Wasserstein distance, Robust gen-

1 | INTRODUCTION

1.1 | Related literature

Optimal transportation (OT) dates back to the work of (Monge, 1781), who considered the problem of finding the optimal way to move given piles of sand to fill up given holes of the same total volume. Monge's problem remained open until it was revisited and solved by Kantorovich, who characterized the optimal transportation plan in relation to the economic problem of optimal allocation of resources. We refer to (Villani, 2009), (Santambrogio, 2015) for a book-length presentation in mathematics, to (Peyré and Cuturi, 2019) for a data science view, and to (Panaretos and Zemel, 2020) for a statistical perspective.

Within the OT setting, the Wasserstein distance ($W_p, p \geq 1$) is defined as the minimum cost of transferring the probability mass from a source distribution to a target distribution, for a given transfer cost function. Nowadays, OT theory and W_p are of fundamental importance for many scientific areas, including statistics, econometrics, information theory and machine learning. For instance, in machine learning, OT based inference is a popular tool in generative models; see e.g. (Arjovsky et al., 2017), (Genevay et al., 2018), (Nietert et al., 2022) and references therein. Similarly, OT techniques are widely-applied for problems related to domain adaptation; see (Courty et al., 2014) and (Balaji et al., 2020). In statistics and econometrics, estimation methods based on W_p are related to minimum distance estimation (MDE); see (Bassetti et al., 2006) and (Bernton et al., 2019). MDE is a research area in continuous development and the estimators based on this technique are called minimum distance estimators. They are linked to M-estimators and may be robust. We refer to (Hampel et al., 1986), (Van der Vaart, 2000) and (Basu et al., 2011) for a book-length statistical introduction. As far as econometrics is concerned, we refer to (Kitamura and Stutzer, 1997) and (Hayashi, 2011). The extant approaches for MDE are usually defined via Kullback-Leibler (KL) divergence, Cressie-Read power divergence, total variation (TV) distance, Hellinger distance, to mention a few. Some of those approaches (e.g. Hellinger distance) require kernel density estimation of the underlying probability density function. Some others (e.g. KL divergence) have poor performance when the considered distributions do not have a common support. Compared to other notions of distance or divergence, W_p avoids the mentioned drawbacks, e.g. it can be applied when the considered distributions do not share the same support and inference is conducted in a generative fashion—namely, a sampling mechanism is considered, in which non-linear functions map a low dimensional latent random vector to a larger dimensional space; see (Genevay et al., 2018). This explains why W_p is a popular inference tool; see e.g. (Courty et al., 2014; Frogner et al., 2015; Kolouri et al., 2017; Carriere et al., 2017; Peyré and Cuturi, 2019); see (Hallin, 2022) and (La Vecchia et al., 2022) for a recent overview of technology transfers among different areas.

In spite of their appealing theoretical properties, OT and W_p have two important inference aspects to deal with: (i) implementation cost and (ii) lack of robustness. (i) The computation of the solution to OT problem and of the computation of W_p can be demanding. To circumvent this problem, (Cuturi, 2013) proposes the use of an entropy regularized version of the OT problem. The resulting divergence is called Sinkhorn divergence; see (Amari et al., 2018) for further details. (Genevay et al., 2018) illustrate the use of this divergence for inference in generative models, via MDE. The proposed estimation method is attractive in terms of performance and speed of computation. However (to the best of our knowledge) the statistical guarantees of the minimum Sinkhorn divergence estimators have not been provided. (ii) Robustness is becoming a crucial aspect for many complex statistical and machine learning applications. The reason is essentially two-fold. First, large datasets (e.g. records of images or large-scale data collected over the internet) can be cluttered with outlying values (e.g. due to recording device failures). Second, every statistical model

represents only an approximation to reality. Thus, the need for inference procedures that remain informative even in the presence of small deviations from the assumed statistical (generative) model is in high demand, in statistics and in machine learning. Robust statistics deals with this problem. We refer to (Hampel et al., 1986) and (Ronchetti and Huber, 2009) for a book-length discussion on robustness principles.

Other papers have already mentioned the robustness issues of W_p and OT; see e.g. (Alvarez-Esteban et al., 2008), (Hallin et al., 2022), (Hallin et al., 2023), (Hallin and Liu, 2023) (del Barrio et al., 2022), (Mukherjee et al., 2021), (Yatracos, 2022) and (Ronchetti, 2022). In the OT literature, (Chizat, 2017) proposes to deal with outliers via unbalanced OT, namely allowing the source and target distribution to be non-standard probability distributions. This sensitivity is an undesired consequence of exactly satisfying the marginal constraints in OT transportation problem: intuitively, using Monge's formulation, the piles of sand and the holes need to have the same total volume, thus the optimal transportation plan is affected by the outlying values inducing a high transportation cost. Relaxing the exact marginal constraints in a way that the transportation plan does not assign large weights to outliers yields the unbalanced OT theory: a mass variation is allowed in the OT problem (intuitively, the piles of sand and the holes do not need to have the same volume) and outlying values can be ignored by the transportation plan. Working along the lines of this approach, (Balaji et al., 2020) define a robust optimal transportation (ROBOT) problem and study its applicability to machine learning. In the same spirit, starting from OT formulation with a regularization term which relaxes the marginal constraints, (Mukherjee et al., 2021) derive a novel robust OT problem, that they label as ROBOT and they apply it to machine learning. Moreover, building on the Minimum Kantorovich Estimator (MKE) of (Bassetti et al., 2006), Mukherjee et al. propose a class of estimators obtained through minimizing the ROBOT and illustrate numerically their performance. In the same spirit, recently, (Nietert et al., 2022) defined a notion of outlier-robust Wasserstein distance which allows for an ϵ -percentage of outlier mass to be removed from contaminated distributions.

1.2 | Our contributions

Working at the intersection between mathematics, probability, statistics, machine learning and econometrics, we study the theoretical, methodological and computational aspects of ROBOT. We review, in a unified perspective, the key aspects of OT and ROBOT, while complementing the extant ROBOT results with our new methodological findings. Our contributions can be summarized as follows.

We prove that the primal ROBOT problem of (Mukherjee et al., 2021) defines a robust distance that we call robust Wasserstein distance $W^{(\lambda)}$, which depends on a tuning parameter λ controlling the level of robustness. We study the measure theoretic properties of $W^{(\lambda)}$, proving that it induces a metric space whose separability and completeness are proved. Moreover, we derive concentration inequalities for the robust Wasserstein distance and illustrate their practical applicability for the selection of λ in the presence of outliers.

These theoretical (in mathematics and probability) findings are instrumental to another contribution (in statistics and econometrics) of this paper: the development of a novel inferential theory based on robust Wasserstein distance. Making use of convex analysis (Rockafellar and Wets, 2009) and of the techniques in (Bernton et al., 2019), we prove that the minimum robust Wasserstein distance estimators exist (almost surely) and are consistent (at the reference model, possibly misspecified). Numerical experiments show that the robust Wasserstein distance estimators remain reliable even in the presence of local departures from the postulated statistical model. These results complement the methodological investigations already available in the literature on minimum distance estimation (in particular the MKE of (Bassetti et al., 2006) and (Bernton et al., 2019)) and on ROBOT ((Balaji et al., 2020) and (Mukherjee et al., 2021)). Indeed, beside the numerical evidence available in the mentioned papers, there are (to the best of our knowledge) no statistical guarantees on the estimators obtained by the method of (Mukherjee et al., 2021) and (Balaji et al., 2020).

Moreover, our results extend the use of OT to conduct reliable inference on random variables with infinite moments.

Another contribution is the derivation of the dual form of ROBOT for application to machine learning problems. This result not only completes the theory in (Mukherjee et al., 2021), but also provides the stepping stone for the implementation of ROBOT: we explain how our duality can be applied to define robust Wasserstein Generative Adversarial Networks (RWGAN). Numerical exercises provide evidence that RWGAN have better performance than the extant Wasserstein GAN (WGAN) models. Moreover, we illustrate how to use ROBOT for outlier detection in domain adaptation, another popular machine learning problem—because of space constraints, we defer this topic to Appendix 5.4.

Finally, as far as the approach in (Nietert et al., 2022) is concerned, we remark that their notion of robust Wasserstein distance has a similar spirit to our $W^{(\lambda)}$ (both are based on a total variation penalization of the original OT problem, but yield different duality formulations) and, similarly to our results in §5.3, it is proved to be useful in generative adversarial network (GAN). In spite of these similarities, our investigation goes deeper in the probabilistic and statistical aspects related to inference methods based on $W^{(\lambda)}$. With this regard, some key difference between (Nietert et al., 2022) and our paper are that Nietert et al.: (a) do not study concentration inequalities for their distance; (b) do not provide indications on how to select and control the ϵ -percentage of outlier mass ignored by the OT solution—for a numerical illustration of this aspect in the setting of GAN, see Section 5.3; (c) do not define the estimators related to the minimization of their outlier-robust Wasserstein distance; (d) use regularization in the dual and a constraint in the primal, whereas ROBOT does the reverse: this simplifies our derivation of the dual problem. In the next sections we take care of all these and other aspects for the ROBOT, providing the theoretical and the applied statistician with a novel toolkit for robust inference via OT.

During the review process of our paper, a referee brought to our attention that, a few weeks after our submission to arXiv, an independent work by (Nietert et al., 2023) was also posted. In their work, they define a class of robust estimators based on the minimization of a partial Wasserstein distance, propose a dual formula for the partial Wasserstein distance, and explore its application to the WGAN.

2 | OPTIMAL TRANSPORT

2.1 | Classical OT

We recall briefly the fundamental notions of OT as formulated by Kantorovich; more details are available in Appendix A. Let $\mathcal{X}$ and $\mathcal{Y}$ denote two Polish spaces, and let $\mathcal{P}(\mathcal{X})$ represent the set of all probability measures on $\mathcal{X}$. Let $\Pi(\mu, \nu)$ denote the set of all joint probability measures of $\mu \in \mathcal{P}(\mathcal{X})$ and $\nu \in \mathcal{P}(\mathcal{Y})$. Kantorovich's problem aims at finding a joint distribution $\pi \in \Pi(\mu, \nu)$ which minimizes the expectation of the coupling between X and Y in terms of the cost function c . This problem can be formulated as

$$\inf \left\{ \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\pi(x, y) : \pi \in \Pi(\mu, \nu) \right\}. \quad (1)$$

A solution to Kantorovich's problem (KP) (1) is called an optimal transport plan. Note that the problem is convex, and its solution exists under some mild assumptions on c , e.g., lower semicontinuous; see e.g. (Villani, 2009, Chapter 4).

KP as in (1) has a dual form, which is related to Kantorovich dual (KD) problem

$$\sup \left\{ \int_{\mathcal{Y}} \phi d\nu - \int_{\mathcal{X}} \psi d\mu : \phi \in C_b(\mathcal{Y}), \psi \in C_b(\mathcal{X}) \right\} \quad \text{s.t.} \quad \phi(y) - \psi(x) \leq c(x, y),$$

for $\forall(x, y)$, where $C_b(\mathcal{X})$ is the set of bounded continuous functions on $\mathcal{X}$. According to Thm. 5.10 in (Villani, 2009), if function c is lower semicontinuous, there exists a solution to the dual problem such that the solutions to KD and KP coincide (no duality gap). In this case, the solution is

$$\phi(y) = \inf_{x \in \mathcal{X}} [\psi(x) + c(x, y)] \quad \text{and} \quad \psi(x) = \sup_{y \in \mathcal{Y}} [\phi(y) - c(x, y)],$$

where the functions ϕ and ψ are called c -concave and c -convex, respectively, and ϕ (resp. ψ) is called the c -transform of ψ (resp. ϕ). When c is a metric on $\mathcal{X}$, OT problem becomes $\sup \left\{ \int_{\mathcal{X}} \psi d\nu - \int_{\mathcal{X}} \psi d\mu : \psi \text{ is 1-Lipschitz continuous} \right\}$, which is the Kantorovich-Rubenstein duality. Now, let $(\mathcal{X}, d)$ denote a complete metric space equipped with a metric $d : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$, and let μ and ν be two probability measures on $\mathcal{X}$. Solving the optimal transport problem in (1), with the cost function $c(x, y) = d^p(x, y)$, introduces a distance, called Wasserstein distance of order p ($p \geq 1$):

$$W_p(\mu, \nu) = \left(\inf_{\pi \in \Pi(\mu, \nu)} \int_{\mathcal{X} \times \mathcal{X}} (d(x, y))^p d\pi(x, y) \right)^{1/p}. \quad (2)$$

With regard to (2), we emphasize that the ability to lift the ground distance $d(x, y)$ is one of the perks of W_p and it makes it a suitable tool in statistics and machine learning; see Remark 2.18 in (Peyré and Cuturi, 2019). Interestingly, this desirable feature becomes a negative aspect as far as robustness is concerned. Intuitively, OT embeds the distributions geometry: when the underlying distribution is contaminated by outliers, the marginal constraints force OT to transport outlying values, inducing an undesirable extra cost, which in turns entails large changes in the W_p . More in detail, let δ_{x_0} be point mass measure centered at x_0 (a point in the sample space), and $d(x, y)$ be the ground distance. Then, we introduce the $O \cup I$ framework, in which the majority of the data $(X_i)_{i \in I}$ are inliers, that is i.i.d. informative data with distribution μ , whilst a few data $(X_i)_{i \in O}$ are outliers, meaning that they are not i.i.d. with distribution μ . We refer to (Lecué and Lerasle, 2020) (and references therein) for more detail on $O \cup I$ and for discussion on its connection to Huber gross-error neighbourhood. Thus, for $\varepsilon > 0$, we have $W_p(\mu, (1 - \varepsilon)\mu + \varepsilon\delta_{x_0}) \rightarrow \infty$, as $x_0 \rightarrow \infty$. Thus, W_p can be arbitrarily inflated by a small number of large observations and it is sensitive to outliers. For instance, let us consider W_p , with $p = 1, 2$. In Figure 1(a) and 1(b), we display the scatter plot for two bivariate distributions: the distribution in panel (b) has some abnormal points (contaminated sample) compared to the distribution in panel (a) (clean sample). Although only a small fraction of the sample data is contaminated, both W_1 and W_2 display a large increase. Anticipating some of the results that we will derive in the next pages, we compute our $W^{(\lambda)}$ (see §3.1) and we notice that its value remains roughly the same in both panels. We refer to §5.1 for additional insights on the resistance to outliers of $W^{(\lambda)}$. The code needed to replicate the numerical exercise in Figure 1 and the codes needed for all the other numerical experiments discussed in the paper, are available at the GitHub repository of the first author (<https://github.com/mayiming24/Robust-optimal-transportation.git>).

2.2 | Robust OT (ROBOT)

We recall the primal formulation of ROBOT problem, as defined in (Mukherjee et al., 2021), who introduce a modified version of Kantorovich OT problem and work on the notion of unbalanced OT. Specifically, they consider a

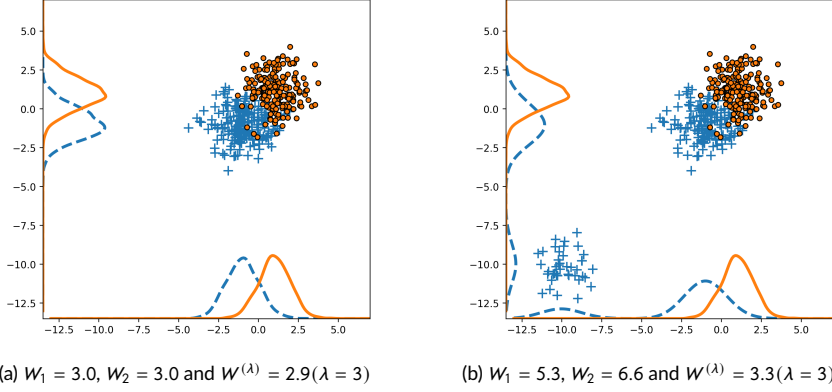


FIGURE 1 Wasserstein distance (W_1 and W_2) and robust Wasserstein distance ($W^{(\lambda)}$, with $\lambda = 3$) between two bivariate distributions. The scatter plot of data in panel (a) represents a sample from the reference model: cross points (blue) are generated from $\mathcal{N}\left(\begin{pmatrix} -1 \\ 1 \end{pmatrix}, I_2\right)$, points (orange) are generated from $\mathcal{N}\left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}, I_2\right)$. The plot in panel (b) contains some outliers: cross points (blue) are generated from $0.8\mathcal{N}\left(\begin{pmatrix} -1 \\ 1 \end{pmatrix}, I_2\right) + 0.2\mathcal{N}\left(\begin{pmatrix} -9 \\ 9 \end{pmatrix}, I_2\right)$, points (orange) are generated from $\mathcal{N}\left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}, I_2\right)$. The marginal distributions are plotted on the x - and y -axis.

TV-regularized OT re-formulation

$$\begin{aligned}
 & \min_{\pi, s} \iint c(x, y) \pi(x, y) dx dy + \lambda \|s\|_{TV} \\
 \text{s.t. } & \int \pi(x, y) dy = \mu(x) + s(x) \geq 0 \\
 & \int \pi(x, y) dx = \nu(y) \\
 & \int s(x) dx = 0,
 \end{aligned} \tag{3}$$

where $\lambda > 0$ is a regularization parameter. In (3), the original source measure μ is modified by adding s (nevertheless, the first and last constraints ensure that $\mu + s$ is still a valid probability measure). Intuitively, having $\mu(x) + s(x) = 0$ means that $x \in \mathcal{X}$ has strong impact on the OT problem and hence can be labelled as an outlier. The outlier is eliminated from the sample, since the probability measure $\mu + s$ at this point is zero.

(Mukherjee et al., 2021) prove that a simplified, computationally efficient formulation equivalent to (3) is

$$\inf \left\{ \int_{\mathcal{X} \times \mathcal{X}} c_\lambda(x, y) d\pi(x, y) : \pi \in \Pi(\mu, \nu) \right\}, \tag{4}$$

which has a functional form similar to (1), but the cost function c is replaced by the trimmed cost function $c_\lambda = \min\{c, 2\lambda\}$ that is bounded from above by 2λ . Following (Mukherjee et al., 2021), we call formulations (3) and (4) ROBOT. Beside the primal formulation, in the following theorem we drive the dual form of the ROBOT, which is not available in (Mukherjee et al., 2021). We need this duality for the development our inferential approach and, for ease of reference, we state it in form of theorem—the proof can be obtained from Thm. 5.10 in (Villani, 2009) and it is available in Appendix B.

Theorem 1 Let μ, ν be probability measures in $\mathcal{P}(\mathcal{X})$, with $c_\lambda(x, y)$ as in (6). Then the c -transform of a c -convex function $\psi(x)$ is itself, i.e. $\psi^c(x) = \psi(x)$. Moreover, the dual form of ROBOT is related to the Kantorovich potential ψ , which is a

solution to

$$\sup \left\{ \int_{\mathcal{X}} \psi d\mu - \int_{\mathcal{X}} \psi d\nu : \psi \in C_b(\mathcal{X}) \right\}, \quad (5)$$

where ψ satisfies $|\psi(x) - \psi(y)| \leq d(x, y)$ and $\text{range}(\psi) \leq 2\lambda$.

Note that the difference between the dual form of OT and that of ROBOT is that the latter imposes a restriction on the range of ψ . Thanks to this, we may think of using the ROBOT in theoretical proofs and inference problems where the dual form of OT is already applied, with the advantage that ROBOT remains stable even in the presence of outliers. In the next sections, we illustrate these ideas and we start the construction of our inference methodology by proving that to ROBOT is associated a robust distance, which will be the pivotal tool of our approach.

Equation (5) is closely related to the bounded-Lipschitz metric in Dudley (1969) defined as

$$d_{BL}(\mu, \nu) = \sup_{\psi} \left\{ \int_{\mathcal{X}} \psi d\mu - \int_{\mathcal{X}} \psi d\nu : \|\psi\|_{BL} \leq 1 \right\},$$

where the bounded-Lipschitz norm of ψ is

$$\|\psi\|_{BL} := \|\psi\|_{\infty} + \|\psi\|_L, \quad \|\psi\|_L := \sup_{x, y \in \mathcal{X}, x \neq y} \frac{|\psi(x) - \psi(y)|}{d(x, y)},$$

denotes the Lipschitz norm. Note that the introduction of λ rather than a constant on $\text{range}(\psi)$ allows us to control the level of truncation in the optimal transport and thus the robustness of our estimators defined in Section 4.

3 | ROBUST WASSERSTEIN DISTANCE $W^{(\lambda)}$

3.1 | Key notions

Setting $c(x, y) = d(x, y)$, (2) implies that the corresponding minimal cost $W_1(\mu, \nu)$ is a metric between two probability measures μ and ν . A similar result can be obtained also in the ROBOT setting of (4), if the modified cost function takes the form

$$c_{\lambda}(x, y) := \min \{d(x, y), 2\lambda\}. \quad (6)$$

Lemma 2 Let $d(\cdot, \cdot)$ denote a metric on $\mathcal{X}$. Then $c_{\lambda}(x, y)$ in (6) is also a metric on $\mathcal{X}$.

Based on Lemma 2, we define the robust Wasserstein distance and prove that it is a metric on $\mathcal{P}(\mathcal{X})$.

Theorem 3 For $c_{\lambda}(x, y)$, $x, y \in \mathcal{X}$, defined in (6),

$$W^{(\lambda)}(\mu, \nu) := \inf_{\pi \in \Pi(\mu, \nu)} \left\{ \int_{\mathcal{X} \times \mathcal{X}} c_{\lambda}(x, y) d\pi(x, y) \right\} \quad (7)$$

is a metric on $\mathcal{P}(\mathcal{X})$, and we call it robust Wasserstein distance.

It is worth noting that W_p is well defined only if probability measures have finite p -th order moments. In contrast, thanks to the boundedness of the c_λ , our $W^{(\lambda)}$ in (7) can be applied to all probability measures, even those with infinite moments of any order. Making use of $W^{(\lambda)}$ we introduce the following

Robust Wasserstein space A robust Wasserstein space is an ordered pair $(\mathcal{P}(X), W^{(\lambda)})$, where $\mathcal{P}(X)$ is the set of all probability measures on a complete separable metric space X , and $W^{(\lambda)}$ defined in Thm. 3 is a (finite) distance on $\mathcal{P}(X)$.

3.2 | Some measure theoretic properties

In this section we state, briefly, some preliminary results. The proofs (which are based on (Villani, 2009)) of these results are available in Appendix B, to which we refer the reader interested in the mathematical details. Here, we provide essentially the key aspects, which we prefer to state in the form of theorems and corollary for the ease of reference in the theoretical developments available in the next sections.

Equipped with Thm. 3 and Definition 3.1, it is interesting to connect W_1 to $W^{(\lambda)}$, via the characterization of its limiting behavior as $\lambda \rightarrow \infty$. Since $W^{(\lambda)}$ is continuous and monotonically increasing with respect to λ , one can prove (via dominated convergence theorem) that its limit coincides with the Wasserstein distance of order $p = 1$. Thus:

Theorem 4 For any probability measures μ and ν in $\mathcal{P}(X)$, $W^{(\lambda)}$ is continuous and monotonically non-decreasing with respect to $\lambda \in [0, \infty)$. Moreover, if $W_1(\mu, \nu)$ exists, we have

$$\lim_{\lambda \rightarrow \infty} W^{(\lambda)}(\mu, \nu) = W_1(\mu, \nu).$$

Thm. 4 has a two-fold implication: first, $W^{(\lambda)} \leq W_1$; second, for large values of the regularization parameter, $W^{(\lambda)}$ and W_1 behave similarly. $W^{(\lambda)}$ and W_1 behave similarly. Additionally, when d is the Euclidean distance, we notice that quantitative bounds on the difference between $W_1(\mu, \nu)$ and $W^{(\lambda)}(\mu, \nu)$ can easily be obtained if μ and ν have finite moments $m_\mu = (\mathbb{E}_{y \sim \nu}[|Y|^p])^{1/p}$ of order $p > 1$, respectively, and $m_\nu = (\mathbb{E}_{x \sim \mu}[|X|^p])^{1/p}$. Indeed, in this case, using that $1\{x > t\} \leq (x/t)^{p-1}$ for any $x, t > 0$, we have, using Thm. 3 and denoting by π_λ an optimal coupling for $W^{(\lambda)}$:

$$\begin{aligned} W_1(\mu, \nu) - W^{(\lambda)}(\mu, \nu) &\leq \int \|x - y\| - \min(\|x - y\|, 2\lambda) d\pi_\lambda(x, y) \\ &\leq \int \|x - y\| 1\{|x - y| > 2\lambda\} d\pi_\lambda(x, y) \\ &\leq \int \frac{\|x - y\|^p}{(2\lambda)^{p-1}} d\pi_\lambda(x, y) \\ &\leq \frac{m_\mu^p + m_\nu^p}{\lambda^{p-1}}. \end{aligned}$$

The first inequality follows from the definition of W_1 , while the last inequality holds due to the inequality

$$(a + b)^p \leq 2^{p-1}(a^p + b^p).$$

This provides a quantitative bound comparing $W_1(\mu, \nu)$ and $W^{(\lambda)}(\mu, \nu)$ as a function of λ and the input measures. Moreover, the last inequality illustrates that the values $\lambda \geq 1$ imply tighter upper bounds on the difference between W_1 and $W^{(\lambda)}$. In the rest of the paper we focus on those values. Another important property of $W^{(\lambda)}$ is that weak convergence is entirely characterized by convergence in the robust Wasserstein distance. More precisely, we have

Theorem 5 Let (X, d) be a Polish space. Then $W^{(\lambda)}$ metrizes the weak convergence in $\mathcal{P}(X)$. In other words, if $(\mu_k)_{k \in \mathbb{N}}$ is a sequence of measures in $\mathcal{P}(X)$ and μ is another measure in $\mathcal{P}(X)$, then $(\mu_k)_{k \in \mathbb{N}}$ converges weakly in $\mathcal{P}(X)$ to μ and $W^{(\lambda)}(\mu_k, \mu) \rightarrow 0$ are equivalent.

The next result follows immediately from Theorem 5.

Corollary 6 (Continuity of $W^{(\lambda)}$) For a Polish space (X, d) , suppose that μ_k (resp. ν_k) converges weakly to μ (resp. ν) in $\mathcal{P}(X)$ as $k \rightarrow \infty$. Then

$$W^{(\lambda)}(\mu_k, \nu_k) \rightarrow W^{(\lambda)}(\mu, \nu).$$

Finally, we prove the separability and completeness of robust Wasserstein space, when X is separable and complete. To this end, we state

Theorem 7 Let X be a complete separable metric space. Then the space $\mathcal{P}(X)$, metrized by the robust Wasserstein distance $W^{(\lambda)}$, is complete and separable. Moreover, the set of finitely supported measures with rational coefficients is dense in $\mathcal{P}(X)$.

We remark that our dual form in Thm. 1 unveils a connection between $W^{(\lambda)}$ and the Bounded Lipschitz (BL) metric discussed in (Dudley, 1969, p.41 and p.42): the two metrics coincide for a suitable selection of λ . With this regard, we emphasize that, since the BL metrizes the weak-star topology, Thm. 3 and Thm. 5 are in line with the results in Section 3 of (Dudley, 1969).

3.3 | Some novel probabilistic aspects

The previous results offer the mathematical toolkit needed to derive some concentration inequalities for $W^{(\lambda)}$ building on the extant results for W_1 . To illustrate this aspect, let us consider the case of Talagrand transportation inequality T_p . We recall that using W_p as a distance between probability measures, transportation inequalities state that, given $\alpha > 0$, a probability measure μ on X satisfies $T_p(\alpha)$ if the inequality

$$W_p(\mu, \nu) \leq \sqrt{2H(\mu|\nu)/\alpha}$$

holds for any probability measure ν , with $H(\mu|\nu)$ being the Kullback-Leibler divergence. For $p = 1$, it has been proven that $T_1(\alpha)$ is equivalent to the existence of a square-exponential moment; see e.g. (Bolley et al., 2007). Now, note that, from Thm. 4, we have $W^{(\lambda)} \leq W_1$: we can immediately claim that if $T_1(\alpha)$ holds, then we have

$$W^{(\lambda)}(\mu, \nu) \leq \sqrt{2H(\mu|\nu)/\alpha}.$$

Thus, we state

Theorem 8 Let μ be a probability measure on $\mathbb{R}^d$, which satisfies a $T_1(\alpha)$ inequality, and $\{X_1, \dots, X_n\}$ be a random sample of independent variables, all distributed according to μ ; let also $\hat{\mu}_n := n^{-1} \sum_{i=1}^n \delta_{X_i}$ be the associated empirical measure, where δ_x is the Dirac distribution with mass on $x \in \mathbb{R}^d$. Then, for any $d' > d$ and $\alpha' < \alpha$, there exists some constant n_0 , depending only on α' , d' and some square-exponential moment of μ , such that for any $\varepsilon > 0$ and

$$n \geq n_0 \max\left(\varepsilon^{-(d'+2)}, 1\right),$$

we have

$$\mathbb{P} \left[W^{(\lambda)}(\mu, \hat{\mu}_n) > \varepsilon \right] \leq \exp\{-0.5\alpha' n \varepsilon^2\}.$$

The proof follows along the lines as the proof of Thm. 2.1 in Bolley et al. combined with Theorem 1 which yields

$$\exp\{-0.5\alpha' 2n \varepsilon^2\} \geq \mathbb{P} [W_1(\mu, \hat{\mu}_n) > \varepsilon] \geq \mathbb{P} \left[W^{(\lambda)}(\mu, \hat{\mu}_n) > \varepsilon \right]. \quad (8)$$

We remark that condition $n \geq n_0 \max(\varepsilon^{-(d'+2)}, 1)$ implies that Thm. 8 has an asymptotic nature. Nevertheless, Bolley et al. prove that

$$\mathbb{P} [W_1(\mu, \hat{\mu}_n) > \varepsilon] \leq C(\varepsilon) \exp\{-0.5\alpha' n \varepsilon^2\}$$

holds for any n with $C(\varepsilon)$ being a (large) constant depending on ε . The argument in (8) implies

$$\mathbb{P} \left[W^{(\lambda)}(\mu, \hat{\mu}_n) > \varepsilon \right] \leq C(\varepsilon) \exp\{-0.5\alpha' n \varepsilon^2\}$$

for any $n \in \mathbb{N}$. This yields a concentration inequality which is valid for any sample size.

We obtained the results in Thm. 8 using an upper bound on W_1 . The simplicity of our argument comes with a cost: for Thm. 8 to hold, the underlying measure μ needs to satisfy some conditions required for W_1 , like e.g. some moment restrictions. To obtain bounds that do not need these stringent conditions, in the next theorem we derive novel non asymptotic upper bounds working directly on $W^{(\lambda)}$. The techniques that we apply to obtain our new results put under the spotlight the role of λ to control the concentration in the $O \cup I$ framework.

Theorem 9 Assume that $X_1, \dots, X_n$ are generated according to the $O \cup I$ setting and let μ denote the common distribution of the inliers. Let $\hat{\mu}_n^{(I)} = |I|^{-1} \sum_{i \in I} \delta_{X_i}$ denote the empirical distribution based on inliers. Then, for any $t > 0$ and, when $O = \emptyset$, we have

$$\mathbb{P} \left(|W^{(\lambda)}(\mu, \hat{\mu}_n^{(I)}) - \mathbb{E}[W^{(\lambda)}(\mu, \hat{\mu}_n^{(I)})]| > \sigma \sqrt{\frac{2t}{|I|}} + \frac{4\lambda t}{|I|} \right) \leq 2 \exp(-t), \quad (9)$$

where the variance term is

$$\sigma^2 = |I|^{-1} \sum_{i \in I} \mathbb{E}[\min(d^2(X_i, X'_i), (2\lambda)^2)]$$

and X'_i is an independent copy of X_i . When $O \neq \emptyset$, we have that, for any $t > 0$,

$$\mathbb{P} \left(\left| W^{(\lambda)}(\mu, \hat{\mu}_n) - \frac{|I|}{n} \mathbb{E}[W^{(\lambda)}(\mu, \hat{\mu}_n^{(I)})] \right| > \sigma \sqrt{\frac{|I|}{n}} \sqrt{\frac{2t}{n}} + \frac{4\lambda t}{n} + \frac{2\lambda|O|}{n} \right) \leq 2 \exp(-t) \quad (10)$$

Some remarks are in order. (i) In the case where $O = \emptyset$, (9) holds without any assumption on μ as the variance term σ^2 is always bounded, even without assuming a finite second moment of X_i . This strongly relaxes the exponential square moment assumption required for $W_1(\mu, \hat{\mu}_n)$ in Thm. 8.

(ii) (10) illustrates that the presence of contamination entails the extra term $2\lambda|O|/n$: one can make use of the

tuning parameter λ to mitigate the influence of outlying values.

(iii) In (10), it is always possible to replace the term

$$|I|E[W^{(\lambda)}(\mu, \hat{\mu}_n^{(I)})]/n$$

by either $E[W^{(\lambda)}(\mu, \hat{\mu}_n)]$ or $E[W^{(\lambda)}(\mu, \hat{\mu}_n')]$, where $\mu'_n = n^{-1} \sum_{i=1}^n \delta_{X'_i}$ and $X'_1, \dots, X'_n$ are i.i.d. with common distribution μ . This can be proved noticing that it is possible to bound $W^{(\lambda)}(\mu, \hat{\mu}_n)$ exploiting the fact that c_λ is bounded by 2λ , so Thm. 1 implies

$$\frac{|I|}{n} W^{(\lambda)}(\mu, \hat{\mu}_n^{(I)}) - \frac{2\lambda|O|}{n} \leq W^{(\lambda)}(\mu, \hat{\mu}_n) \leq \frac{|I|}{n} W^{(\lambda)}(\mu, \hat{\mu}_n^{(I)}) + \frac{2\lambda|O|}{n} . \quad (11)$$

(iv) Delving more into the link between (10) and (11), consider that, for any $\psi \in C_b(X)$,

$$\begin{aligned} \int_X \psi d\hat{\mu}_n - \int_X \psi d\mu &= \frac{1}{n} \sum_{i=1}^n \left\{ \psi(X_i) - \int_X \psi d\mu \right\} \\ &= \frac{1}{n} \left(\sum_{i \in I} \left\{ \psi(X_i) - \int_X \psi d\mu \right\} + \sum_{i \in O} \left\{ \psi(X_i) - \int_X \psi d\mu \right\} \right) \\ &= \frac{1}{n} \left(|I| \left\{ \int_X \psi d\hat{\mu}_n^{(I)} - \int_X \psi d\mu \right\} + \sum_{i \in O} \left\{ \psi(X_i) - \int_X \psi d\mu \right\} \right) . \end{aligned}$$

As $\text{range}(\psi) \leq 2\lambda$, for any $i \in O$, it holds that

$$\left| \psi(X_i) - \int_X \psi d\mu \right| \leq 2\lambda .$$

Therefore, we have, for any $\psi \in C_b(X)$,

$$\left| \int_X \psi d\hat{\mu}_n - \int_X \psi d\mu \right| - \frac{|I|}{n} \left| \int_X \psi d\hat{\mu}_n^{(I)} - \int_X \psi d\mu \right| \leq \frac{2\lambda|O|}{n} .$$

Taking the suprema over ψ in $C_b(X)$ yields (11). From (11) and the triangular inequality, we have that (adding and subtracting $\frac{|I|}{n} W^{(\lambda)}(\mu, \hat{\mu}_n^{(I)})$)

$$\begin{aligned} \left| W^{(\lambda)}(\mu, \hat{\mu}_n) - \frac{|I|}{n} E[W^{(\lambda)}(\mu, \hat{\mu}_n^{(I)})] \right| &\leq \frac{|I|}{n} \left| W^{(\lambda)}(\mu, \hat{\mu}_n^{(I)}) - E[W^{(\lambda)}(\mu, \hat{\mu}_n^{(I)})] \right| \\ &\quad + \frac{2\lambda|O|}{n} . \end{aligned}$$

Inequality (10) follows by plugging (9) into this last bound. Finally, combining (10) and (11), we obtain

$$\mathbb{P} \left(\left| W^{(\lambda)}(\mu, \hat{\mu}_n) - E[W^{(\lambda)}(\mu, \hat{\mu}_n)] \right| > \sigma \sqrt{\frac{|I|}{n}} \sqrt{\frac{2t}{n}} + \frac{4\lambda t}{n} + \frac{4\lambda|O|}{n} \right) \leq 2 \exp(-t) . \quad (12)$$

Beside the concentrations in Thm. 9, we consider the problem of providing an upper bound to $E[W^{(\lambda)}(\mu, \hat{\mu}_n^{(I)})]$. Following (Boissard and Le Gouic, 2014), we call this quantity the mean rate of convergence. For the sake of exposition, we assume that all data are inliers, that is $X_1, \dots, X_n$ are i.i.d. with common distribution μ —the case with outliers can be obtained using (11). As in Thm. 8, we notice that it is always possible to use $W^{(\lambda)} \leq W_1$ from Thm. 4 and derive

bounds for $E[W^{(\lambda)}(\mu, \hat{\mu}_n)]$ applying the results for $E[W_1(\mu, \hat{\mu}_n)]$; see e.g. (Boissard and Le Gouic, 2014), (Fournier and Guillin, 2015; Lei, 2020a; Fournier, 2022). Beside this option, here we derive novel results for $E[W^{(\lambda)}(\mu, \hat{\mu}_n)]$. In what follows, we assume that μ is supported in $B(0, K)$, the ball in Euclidean distance of $\mathbb{R}^d$ with radius K and that the reference distance is the Euclidean distance in $\mathbb{R}^d$. Then we state:

Theorem 10 *Let μ denote a measure on $B(0, K)$, the ball in Euclidean distance of $\mathbb{R}^d$ with radius K . Let $X_1, \dots, X_n$ denote an i.i.d. sample of μ and $\hat{\mu}_n$ denote the empirical measure. Then, we have*

$$E[W^{(\lambda)}(\mu, \hat{\mu}_n)] \leq \begin{cases} C \sqrt{\frac{K(K \wedge \lambda)}{n}} & \text{when } d = 1, \\ \frac{CK}{\sqrt{n}} \log\left(\frac{\sqrt{n}}{K}\right) & \text{when } d = 2, \\ \frac{CK}{n^{1/d}} & \text{when } d \geq 3. \end{cases}$$

The bound is obtained using (i) the duality formula derived in Thm. 1, (ii) an extension of Dudley's inequality in the spirit of (Boissard and Le Gouic, 2014), and (iii) a computation of the entropy number of a set of 1-Lipschitz functions defined on the ball $B(0, K)$ and taking value in $[-\lambda, \lambda]$. The technical details are available in Appendix B. Here, we remark the key role of λ when $d = 1$ (univariate case), which allows to bound the mean rate of convergence. For $d \geq 2$, our bounds could have been obtained also from (Boissard and Le Gouic, 2014). However, the obtained results can be substantially larger than our upper bounds when $\lambda \ll K$: Thm. 10 provides a refinement (related to the boundedness of c_λ) of the extant bounds.

Moreover, still for $d > 2$, we emphasize that the most important effect on the mean rate of convergence is related to the term in d , which expresses the key role of dimensionality, rather than to the term containing the robustness parameter λ ; see also Proposition 9 in (Nietert et al., 2022) for a similar comment. The technical explanation for this result is in Step 1 – Step 3 of the proof of Thm. 10, as available in the Supplementary Material and to which we refer—in particular, see our discussion on the convergence of the Dudley's series, at the beginning of Step 3. Finally, as remarked by an anonymous referee, we notice that, using Thm. 3.1 in (Lei, 2020b) together with $W^{(\lambda)} \leq W_1$, one may extend Thm. 10 to non-compactly supported measures, when $d > 2$.

4 | INFERENCE VIA MINIMUM $W^{(\lambda)}$ ESTIMATION

4.1 | Estimation method

Let us consider a probability space $(\Omega, \mathcal{F}, P)$. On this probability space, we define random variables taking values on $\mathcal{X} \subset \mathbb{R}^d$, $d \geq 1$ and endowed with the Borel σ -algebra. We observe $n \in \mathbb{N}$ i.i.d. data points $\{X_1, \dots, X_n\}$, which are distributed according to $\mu_\star^{(n)} \in \mathcal{P}(\mathcal{X}^n)$. A parametric statistical model on $\mathcal{X}^n$ is denoted by $\{\mu_\theta^{(n)}\}_{\theta \in \Theta}$ and it is a collection of probability distributions indexed by a parameter θ of dimension d_θ . The parameter space is $\Theta \subset \mathbb{R}^{d_\theta}$, $d_\theta \geq 1$, which is equipped with a distance ρ_Θ . We let $Z_{1:n}$ represent the observations from $\mu_\theta^{(n)}$. For every $\theta \in \Theta$, the sequence $(\hat{\mu}_{\theta,n})_{n \geq 1}$ of random probability measures on $\mathcal{X}$ converges (in some sense) to a distribution $\mu_\theta \in \mathcal{P}(\mathcal{X})$, where

$$\hat{\mu}_{\theta,n} = n^{-1} \sum_{i=1}^n \delta_{Z_i}$$

with $Z_{1:n} \sim \mu_{\theta}^{(n)}$. Similarly, we will assume that the empirical measure $\hat{\mu}_n$ converges (in some sense) to some distribution $\mu_{\star} \in \mathcal{P}(\mathcal{X})$ as $n \rightarrow \infty$. We say that the model is well-specified if there exists $\theta_{\star} \in \Theta$ such that $\mu_{\star} \equiv \mu_{\theta_{\star}}$; otherwise, it is misspecified. Parameters are identifiable: $\theta_1 = \theta_2$ is implied by $\mu_{\theta_1} = \mu_{\theta_2}$.

Our estimation method relies on selecting a parametric model in $\{\mu_{\theta}^{(n)}\}_{\theta \in \Theta}$, which is the closest, in robust Wasserstein distance, to the true model $\mu_{\star}$. Thus, minimum $W^{(\lambda)}$ estimation refers to the minimization, over the parameter $\theta \in \Theta$, of the robust Wasserstein distance between the empirical distribution $\hat{\mu}_n$ and the reference model distribution μ_{θ} . This is similar to the approach described in (Bassetti et al., 2006) and (Bernton et al., 2019), who derive minimum Kantorovich estimators by making use of W_p .

More formally, the minimum robust Wasserstein estimator (MRWE) is defined as

$$\hat{\theta}_n^{\lambda} = \underset{\theta \in \Theta}{\operatorname{argmin}} W^{(\lambda)}(\hat{\mu}_n, \mu_{\theta}). \quad (13)$$

When there is no explicit expression for the probability measure characterizing the parametric model (e.g. in complex generative models, see (Genevay et al., 2018) and § 5.2 for some examples), the computation of MRWE can be difficult. To cope with this issue, in lieu of (13), we propose using the minimum expected robust Wasserstein estimator (MERWE) defined as

$$\hat{\theta}_{n,m}^{\lambda} = \underset{\theta \in \Theta}{\operatorname{argmin}} E_m \left[W^{(\lambda)}(\hat{\mu}_n, \hat{\mu}_{\theta,m}) \right]. \quad (14)$$

where the expectation E_m is taken over the distribution $\mu_{\theta}^{(m)}$. To implement the MERWE one can rely on Monte Carlo methods and approximate numerically $E_m[W^{(\lambda)}(\hat{\mu}_n, \hat{\mu}_{\theta,m})]$. Replacing the robust Wasserstein distance with the W_p in (14), one obtains the minimum Wasserstein estimator (MWE) and the minimum expected Wasserstein estimator (MEWE), studied in (Bernton et al., 2019).

4.2 | Statistical guarantees

Intuitively, the consistency of the MRWE and MERWE can be conceptualized as follows. We expect that the empirical measure converges to $\mu_{\star}$, in the sense that $W^{(\lambda)}(\hat{\mu}_n, \mu_{\star}) \rightarrow 0$ as $n \rightarrow \infty$; see Thm. 5. Therefore, the argmin of $W^{(\lambda)}(\hat{\mu}_n, \mu_{\star})$ should converge to

$$\theta^* = \arg \min W^{(\lambda)}(\mu_{\star}, \mu_{\theta}),$$

assuming its existence and unicity. The same can be said for the minimum of the MERWE, provided that $m \rightarrow \infty$. If the reference parametric model is correctly specified (e.g. no data contamination), θ^* is the limiting object of interest and it is the minimizer of $W^{(\lambda)}(\mu_{\star}, \mu_{\theta})$. In the case of model misspecification (e.g. wrong parametric form of μ_{θ} and/or presence of data contamination), θ^* is not necessarily the parameter that minimizes the KL divergence between the empirical measure and the measure characterizing the reference model. We emphasize that this is at odd with the standard misspecification theory (see e.g. (White, 1982)) and it is due to the fact that we replace the KL divergence (which yields non robust estimators) with our novel robust Wasserstein distance.

To formalize these arguments, we introduce the following set of assumptions, which are standard in the literature on MKE; see (Bernton et al., 2019).

Assumption 11 *The data-generating process is such that $W^{(\lambda)}(\hat{\mu}_n, \mu_{\star}) \rightarrow 0$, P-almost surely as $n \rightarrow \infty$.*

Assumption 12 The map $\theta \mapsto \mu_\theta$ is continuous in the sense that $\rho_\Theta(\theta_n, \theta) \rightarrow 0$ implies that μ_{θ_n} converges to μ_θ weakly as $n \rightarrow \infty$.

Assumption 13 For some $\varepsilon > 0$,

$$B_\star(\varepsilon) = \left\{ \theta \in \Theta : W^{(\lambda)}(\mu_\star, \mu_\theta) \leq \varepsilon_\star + \varepsilon \right\},$$

with $\varepsilon_\star = \inf_{\theta \in \Theta} W^{(\lambda)}(\mu_\star, \mu_\theta)$, is a bounded set.

Then we state the following

Theorem 14 (Existence of MRWE) Under Assumptions 11,12 and 13, there exists a set $\mathcal{A} \subset \Omega$ with $P(\mathcal{A}) = 1$ such that, for all $\omega \in \mathcal{A}$,

$$\inf_{\theta \in \Theta} W^{(\lambda)}(\hat{\mu}_n(\omega), \mu_\theta) \rightarrow \inf_{\theta \in \Theta} W^{(\lambda)}(\mu_\star, \mu_\theta)$$

and there exists $n(\omega)$ such that, for all $n \geq n(\omega)$, the sets

$$\operatorname{argmin}_{\theta \in \Theta} W^{(\lambda)}(\hat{\mu}_n(\omega), \mu_\theta)$$

are non-empty and form a bounded sequence with

$$\limsup_{n \rightarrow \infty} \operatorname{argmin}_{\theta \in \Theta} W^{(\lambda)}(\hat{\mu}_n(\omega), \mu_\theta) \subset \operatorname{argmin}_{\theta \in \Theta} W^{(\lambda)}(\mu_\star, \mu_\theta).$$

To prove Thm. 14 we need to show that the sequence of functions $\theta \mapsto W^{(\lambda)}(\hat{\mu}_n(\omega), \mu_\theta)$ epi-converges (see Definition B in Appendix B) to $\theta \mapsto W^{(\lambda)}(\mu_\star, \mu_\theta)$. The result follows from (Rockafellar and Wets, 2009). We remark that Thm. 14 generalizes the results of (Bassetti et al., 2006) and (Bernton et al., 2019), where the model is assumed to be well specified ((Bassetti et al., 2006)) and moments of order $p \geq 1$ are needed ((Bernton et al., 2019)).

Moving along the same lines as in Thm. 2.1 in (Bernton et al., 2019), we may prove the measurability of MRWE; see Thm. 5 in Appendix A. These results for the MRWE provide the stepping stone to derive similar theorems for the MERWE $\hat{\theta}_{n,m}^\lambda$. To this end, the following assumptions are needed.

Assumption 15 For any $m \geq 1$, if $\rho_\Theta(\theta_n, \theta) \rightarrow 0$, then $\mu_{\theta_n}^{(m)}$ converges to $\mu_\theta^{(m)}$ weakly as $n \rightarrow \infty$.

Assumption 16 If $\rho_\Theta(\theta_n, \theta) \rightarrow 0$, then $E_n W^{(\lambda)}(\mu_{\theta_n}, \hat{\mu}_{\theta_n, n}) \rightarrow 0$ as $n \rightarrow \infty$.

Theorem 17 (Existence of MERWE) Under Assumptions 11,12,13,15 and 16, there exists a set $\mathcal{A} \subset \Omega$ with $P(\mathcal{A}) = 1$ such that, for all $\omega \in \mathcal{A}$,

$$\inf_{\theta \in \Theta} E_{m(n)} \left[W^{(\lambda)}(\hat{\mu}_n(\omega), \hat{\mu}_{\theta, m(n)}) \right] \rightarrow \inf_{\theta \in \Theta} W^{(\lambda)}(\mu_\star, \mu_\theta)$$

and there exists $n(\omega)$ such that, for all $n \geq n(\omega)$, the sets

$$\operatorname{argmin}_{\theta \in \Theta} W^{(\lambda)}(\hat{\mu}_n(\omega), \hat{\mu}_{\theta, m(n)})$$

are non-empty and form a bounded sequence with

$$\limsup_{n \rightarrow \infty} \operatorname{argmin}_{\theta \in \Theta} E_{m(n)} \left[W^{(\lambda)}(\hat{\mu}_n(\omega), \hat{\mu}_{\theta, m(n)}) \right] \subset \operatorname{argmin}_{\theta \in \Theta} W^{(\lambda)}(\mu_*, \mu_\theta).$$

For a generic function f , let us define

$$\varepsilon\text{-argmin}_x f := \left\{ x : f(x) \leq \varepsilon + \inf_x f \right\}.$$

Then, we state the following

Theorem 18 (Measurability of MERWE) Suppose that Θ is a σ -compact Borel measurable subset of $\mathbb{R}^{d_\theta}$. Under Assumption 15, for any $n \geq 1$ and $m \geq 1$ and $\varepsilon > 0$, there exists a Borel measurable function $\hat{\theta}_{n,m} : \Omega \rightarrow \Theta$ that satisfies

$$\hat{\theta}_{n,m}^\lambda(\omega) \in \operatorname{argmin}_{\theta \in \Theta} E_m \left[W^{(\lambda)}(\hat{\mu}_n(\omega), \hat{\mu}_{\theta, m}) \right],$$

if this set is non-empty, otherwise,

$$\hat{\theta}_{n,m}^\lambda(\omega) \in \varepsilon\text{-argmin}_{\theta \in \Theta} E_m \left[W^{(\lambda)}(\hat{\mu}_n(\omega), \hat{\mu}_{\theta, m}) \right].$$

Considering the case where the data and n are fixed, the next result shows that the MERWE converges to the MRWE, as $m \rightarrow \infty$. In the next assumption, the observed empirical distribution is kept fix and $\varepsilon_n = \inf_{\theta \in \Theta} W^{(\lambda)}(\hat{\mu}_n, \mu_\theta)$.

Assumption 19 For some $\varepsilon > 0$, the set

$$B_n(\varepsilon) = \left\{ \theta \in \Theta : W^{(\lambda)}(\hat{\mu}_n, \mu_\theta) \leq \varepsilon_n + \varepsilon \right\}$$

is bounded.

Theorem 20 (MERWE converges to MRWE as $m \rightarrow \infty$) Under Assumptions 12, 15, 16 and 19,

$$\inf_{\theta \in \Theta} E_m \left[W^{(\lambda)}(\hat{\mu}_n, \hat{\mu}_{\theta, m}) \right] \rightarrow \inf_{\theta \in \Theta} W^{(\lambda)}(\hat{\mu}_n, \mu_\theta),$$

and there exists a $\tilde{m}$ such that, for all $m \geq \tilde{m}$, the sets

$$\operatorname{argmin}_{\theta \in \Theta} E_m W^{(\lambda)}(\hat{\mu}_n, \hat{\mu}_{\theta, m})$$

are non-empty and form a bounded sequence with

$$\limsup_{m \rightarrow \infty} \operatorname{argmin}_{\theta \in \Theta} E_m \left[W^{(\lambda)}(\hat{\mu}_n, \hat{\mu}_{\theta, m}) \right] \subset \operatorname{argmin}_{\theta \in \Theta} W^{(\lambda)}(\hat{\mu}_n, \mu_\theta).$$

Remark. Assumption 11 is needed to study the consistency of the estimator at the reference model, namely when $\varepsilon \equiv 0$. This yields the results presented in the theorems of this section. This type of development is in line with the framework typically considered in robust statistics, where asymptotic properties (consistency and asymptotic normality) of M-estimators are conceptualized when the empirical measure converges to the population measure μ^* ;

for a book-length introduction, see e.g. Eq. (2.2) in (Maronna et al., 2006) and (Huber and Ronchetti, 2011), or (Mancini et al., 2005) and (La Vecchia, 2016) for papers containing related discussions. When $\epsilon > 0$, the reference model μ^* remains valid for the majority of the data, whilst the ϵ -percentage of contaminated data behaves differently—Figure 2 provides a graphical illustration, where the blue crosses are far from the bulk of the other blue data points, in the $O \cup I$ setting. The concentration inequalities of Section 4 show that, for finite n , the MRWE reduces the impact of anomalous records by controlling the concentration of $W^{(\lambda)}(\hat{\mu}_n, \mu)$. In Section 6, we elaborate further on that and we explain how one can fine-tune the stability of the MRWE by controlling its bias in the presence of contamination via the tuning constant λ —for a different approach based on asymptotic arguments ($n \rightarrow \infty$), we refer to Section 2.2 in (Nietert et al., 2023), where the authors characterize the information complexity of MDE by the modulus of continuity and build on the results of (Donoho and Liu, 1988) to provide robustness guarantees under TV-norm. Besides these theoretical findings, the Monte Carlo experiments in Sections 5.2 and 5.3 provide numerical evidence that, under various ϵ -specifications and different sample sizes (see Table 1 hereunder), the boundedness of $W^{(\lambda)}$ ensures that the MRWE outperforms significantly its non-robust competitors (whose distribution can be arbitrarily deformed by outlying data points) and it remains informative on the distribution of the bulk of the data.

5 | NUMERICAL EXPERIMENTS

5.1 | Sensitivity to outliers

To complement the insights obtained looking at Figure 1, we build on the notion of sensitive curve, which is an empirical tool that illustrates the stability of statistical functionals; see (Tukey, 1977) and (Hampel et al., 1986) for a book-length presentation. We consider a similar tool to study the sensitivity of W_1 and $W^{(\lambda)}$ in finite samples, both interpreted as functionals of the empirical measure. More in detail, we fix the standard normal distribution as a reference model (for this distribution both $W^{(\lambda)}$ and W_1 are well-defined) and we use it to generate a sample of size $n = 1000$. We denote the resulting sample $\mathbf{X}_n := (X_1, \dots, X_n)$, whose empirical measure is $\hat{\mu}_n$. Then, we replace X_n with a value $x \in \mathbb{R}$ and form a new set of sample points, denoted by $\mathbf{X}_n(x)$, whose empirical measure is $\hat{\mu}_n^{(x)}$. We let T_n represent the robust Wasserstein distance between an n -dimensional sample and $\mathbf{X}_n$. So, $T_n(\mathbf{X}_n)$ is the empirical robust Wasserstein distance between $\mathbf{X}_n$ and its self, thus it is equal to 0, while $T_n(\mathbf{X}_n(x))$ is the empirical robust Wasserstein distance between $\mathbf{X}_n(x)$ and $\mathbf{X}_n$. Finally, for different values of x and λ , we compute (numerically)

$$\Delta(x, W^{(\lambda)}) = n [T_n(X_1, \dots, X_{n-1}, x) - T_n(X_1, \dots, X_n)] = n W^{(\lambda)}(\hat{\mu}_n, \hat{\mu}_n^{(x)}).$$

A similar procedure is applied to obtain $\Delta(x, W_1)$. We display the results in Figure 2. For each value of λ , the plots illustrate that $\Delta(x, W^{(\lambda)})$ first grows and then remains flat (at the value of 2λ) even for very large values of $|x|$. In contrast, for $|x| \rightarrow \infty$, $\Delta(x, W_1)$ diverges, in line with the evidence from (and the comments on) Figure 1. In addition, we notice that as $\lambda \rightarrow \infty$, the plot of $\Delta(x, W^{(\lambda)})$ becomes more and more similar to the one of $\Delta(x, W_1)$: this aspect is in line with Thm. 4 and it is reminiscent of the behaviour of the Huber loss function ((Huber, 1992)) for location estimation, which converges to the quadratic loss (maximum likelihood estimation), as the constant tuning the robustness goes to infinity. However, an important remark is order: the cost c_λ yields, in the language of robust statistics, the so-called “hard rejection”: it bounds the influence of outlying values (to be contrasted with the behavior of Huber loss, which downweights outliers to preserve efficiency at the reference model); see (Ronchetti, 2022) for a similar comment.

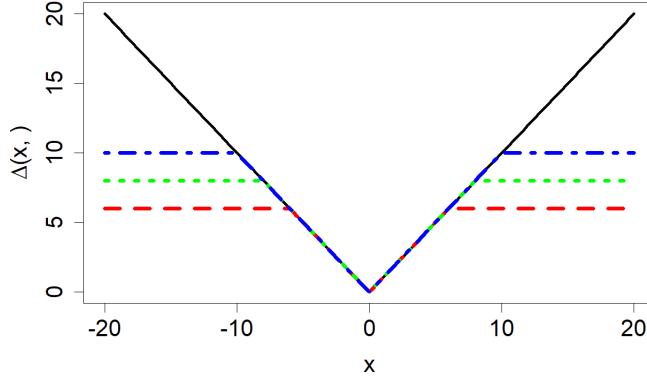


FIGURE 2 The continuous line represents $\Delta(x, W_1)$. The dashed (red), dot-dashed (blue) and dotted (green) line represents $\Delta(x, W^{(\lambda)})$ with $\lambda = 3, 4, 5$ respectively.

5.2 | Estimation of location

Let us consider the statistical problem of inference on univariate location models. We study the robustness of MERWE, comparing its performance to the one of MEWE based on minimizing W_1 , under different degrees of data contamination and for different underlying data generating models (with finite and infinite moments). Before delving into the numerical exercise, let us give some numerical details. We recall that the MERWE aims at minimizing the function

$$\theta \mapsto E_m \left[W^{(\lambda)}(\hat{\mu}_n, \hat{\mu}_{\theta, m}) \right].$$

With the same notation as in §4.1, suppose we generate k replications $Z_{1:m}^{(i)}, i = 1, \dots, k$, independently from a reference model $\mu_\theta^{(m)}$, and let $\hat{\mu}_{\theta, m}^{(i)}$ denote the empirical probability measure of $Z_{1:m}^{(i)}$. Then the loss function

$$L_{\text{MERWE}} = k^{-1} \sum_{i=1}^k W^{(\lambda)}(\hat{\mu}_n, \hat{\mu}_{\theta, m}^{(i)})$$

is a natural estimator of $E_m[W^{(\lambda)}(\hat{\mu}_n, \hat{\mu}_{\theta, m})]$, since the former converges almost surely to the latter as $k \rightarrow \infty$. We note that the algorithmic complexity in m is super-linear while the complexity in k is linear: the incremental cost of increasing k is lower than that one of increasing m . Moreover, a key aspect for the implementation of our estimator is related to the need for specifying λ . In the next numerical exercises we set $\lambda = 5$: this value yields a good performance (in terms of Mean-Squared Error, MSE) of our estimators across the Monte Carlo settings, which consider different models and levels of contamination. In §6, we describe a data driven procedure for selecting λ .

In our numerical investigation, first, we consider a reference model which is the sum of log-normal random variables having finite moments. For a given $L \geq 1$, $\gamma \in \mathbb{R}$ and $\sigma > 0$, we have $X = \sum_{l=1}^L \exp(Z_l)$, where $Z_1, \dots, Z_L$ are sampled from $\mathcal{N}(\gamma, \sigma^2)$ independently. Suppose we are interested in estimating the location parameter γ . We investigate the performance of MERWE and MEWE, under different scenarios: namely, in presence of outliers, for

different sample sizes and contamination values. Specifically, we consider the model

$$X_i^{(n)} = \sum_{l=1}^L \exp(Z_{il}^{(n)}), l = 1, \dots, L, i = 1, \dots, n,$$

where $Z_{il}^{(n)} \sim \mathcal{N}(\gamma, \sigma)$ for $i = 1, \dots, n_1$ (clean part of the sample) and $Z_{il}^{(n)} \sim \mathcal{N}(\gamma + \eta, \sigma)$ for $i = n_1 + 1, \dots, n$ (contaminated part of the sample). Therefore, in each sample, there are $n - n_1$ outliers of size η .

In our simulation experiments, we set $L = 10$, $\gamma = 0$ and $\sigma = 1$. To implement the MERWE and MEWE of γ , we choose $m = 1000$, $k = 20$ and $\lambda = 5$. The bias and MSE, based on 1000 replications, of the estimators are displayed in Table 1, for various sample sizes n , different contamination size η and proportion of contamination ε . The table illustrates the superior performance (both in terms of bias and MSE) of the MERWE with respect to the MEWE. In small samples ($n = 100$), the MERWE has smaller bias and MSE than the MEWE, in all settings. Similar results are available in moderate and large sample size ($n = 200$ and $n = 1000$). Interestingly, for $n = 1000$, MERWE and MEWE have similar performance when $\varepsilon = 0$ (no contamination), whilst the MERWE still has smaller MSE for $\varepsilon > 0$. This implies that the MERWE maintains good efficiency with respect to MEWE at the reference model.

	$n = 100$				$n = 200$				$n = 1000$			
	BIAS		MSE		BIAS		MSE		BIAS		MSE	
	MERWE	MEWE	MERWE	MEWE	MERWE	MEWE	MERWE	MEWE	MERWE	MEWE	MERWE	MEWE
$\varepsilon = 0.1, \eta = 1$	0.049	0.092	0.003	0.010	0.042	0.093	0.002	0.012	0.037	0.086	0.002	0.008
$\varepsilon = 0.1, \eta = 4$	0.035	0.090	0.002	0.012	0.029	0.097	0.001	0.016	0.013	0.100	≈ 0	0.018
$\varepsilon = 0.2, \eta = 1$	0.071	0.157	0.008	0.028	0.086	0.178	0.009	0.033	0.081	0.172	0.007	0.031
$\varepsilon = 0.2, \eta = 4$	0.046	0.204	0.003	0.045	0.035	0.203	0.002	0.043	0.017	0.195	≈ 0	0.038
$\varepsilon = 0$	0.036	0.034	0.002	0.002	0.022	0.022	0.001	0.001	0.012	0.010	≈ 0	≈ 0

TABLE 1 The bias and MSE of the MERWE and MEWE of γ for different n , ε and η .

Now, let us turn to the case of random variables having infinite moments. (Yatracos, 2022) recently illustrates that minimum Wasserstein distance estimators do not perform well for heavy-tailed distributions and suggests to avoid the use of minimum Wasserstein distance inference when the underlying distribution has infinite p -th moment, with $p \geq 1$. The theoretical developments of Section 3.1 show that our MERWE does not suffer from the same criticism. In the next MC experiment, we illustrate the good performance of MERWE when the data generating process does not admit finite first moment. We consider the problem of estimating the location parameters for a symmetric α -stable distribution; see e.g. (Samorodnitsky and Taqqu, 2017) for a book-length presentation. A stable distribution is characterized by four parameter $(\alpha, \beta, \gamma, \delta)$: α is the index parameter, β is the skewness parameter, γ is the scale parameter and δ is the location parameter. It is worth noting that stable distributions have undefined variance for $\alpha < 2$, and undefined mean for $\alpha \leq 1$. We consider three parameters setting:

- (1) $(\alpha, \beta, \gamma, \delta) = (0.5, 0, 1, 0)$, which represents a heavy-tailed distribution without defined mean;
- (2) $(\alpha, \beta, \gamma, \delta) = (1, 0, 1, 0)$ which is the standard Cauchy distribution, having undefined moment of order $p \geq 1$;
- (3) $(\alpha, \beta, \gamma, \delta) = (1.1, 0, 1, 0)$ representing a distribution having a finite mean.

In each MC experiment, we estimate the location parameter, while the other parameters are supposed to be known. In addition, we consider contaminated heavy-tailed data, where $(1 - \varepsilon)$ proportion of n observations is generated from

α -stable distribution with parameter $(\alpha, \beta, \gamma, \delta)$ and the other ε proportion (outliers) comes from the distribution with parameter $(\alpha, \beta, \gamma, \delta + \eta)$ (η is the size of outliers). We set $m = 1000$, $n = 100$ and $k = 20$ and repeat the experiment 1000 times for each distribution and estimator, and $\lambda = 5$. We display the results in Table 2. For the stable distributions, the MEWE has larger bias and MSE than the ones yielded by the MERWE. This aspect is particularly evident for the distributions with undefined first moment, namely the Cauchy distribution and the stable distribution with $\alpha = 0.5$. These experiments, complementing the ones available in (Yatracos, 2022), illustrate that while MEWE (which is not well-defined in the considered setting) entails large bias and MSE values, MERWE is well-defined and performs well even for stable distributions with infinite moments.

	Cauchy				Stable ($\alpha = 0.5$)				Stable ($\alpha = 1.1$)			
	BIAS		MSE		BIAS		MSE		BIAS		MSE	
	MERWE	MEWE	MERWE	MEWE	MERWE	MEWE	MERWE	MEWE	MERWE	MEWE	MERWE	MEWE
$\varepsilon = 0.1, \eta = 1$	0.084	1.531	0.011	3.628	0.088	3.179	0.011	13.731	0.090	0.658	0.011	1.030
$\varepsilon = 0.1, \eta = 4$	0.205	1.529	0.047	3.656	0.164	3.174	0.034	13.706	0.207	0.746	0.048	1.051
$\varepsilon = 0.2, \eta = 1$	0.181	1.502	0.038	3.602	0.171	3.155	0.037	12.840	0.181	0.676	0.038	0.942
$\varepsilon = 0.2, \eta = 4$	0.459	1.820	0.224	4.691	0.384	3.140	0.166	12.713	0.485	1.072	0.245	1.802
$\varepsilon = 0$	0.046	1.551	0.003	3.740	0.045	3.118	0.003	12.600	0.042	0.613	0.003	0.894

TABLE 2 The bias and MSE of the MERWE and MEWE of location parameter for various contaminated α -stable distributions. Simulation setting: $m = 1000$, $k = 20$.

5.3 | Generative Adversarial Networks (GAN)

Synthetic data. We propose two RWGAN deep learning models: both approaches are based on ROBOT. The first one is derived directly from the dual form in Thm. 1, while the second one is derived from (3). We compare these two methods with routinely-applied Wasserstein GAN (WGAN) and with the robust WGAN introduced by (Balaji et al., 2020). To provide some details on our new procedures, we recall that GAN is a deep learning method, first proposed by (Goodfellow et al., 2014). It is one of the most popular machine learning approaches for unsupervised learning on complex distributions. GAN includes a generator and a discriminator that are both neural networks and the procedure is based on mutual game learning between them. The generator creates fake samples as well as possible to deceive the discriminator, while the discriminator needs to be more and more powerful to detect the fake samples. The ultimate goal of this procedure is to produce a generator with a great ability to produce high-quality samples just like the original sample. To illustrate the connections with the ROBOT, let us consider the following GAN architecture; more details are available in the Supplementary Material (Appendix D.2). Let $X \sim P_r$ and $Z \sim P_z$, where P_r and P_z denote the distribution of the reference sample and of a random sample which is the input for the generator. Then, denote by G_θ the function applied by generator (it transforms Z to create fake samples, which are the output of a statistical model P_θ , indexed by the finite dimensional parameter θ) and by D_δ the function applied by the discriminator (it is indexed by a finite dimensional parameter δ , which outputs the estimate of the probability that the sample is true). The objective function is

$$\min_{\theta} \max_{\delta} \{E[\log D_\delta(X)] + E[\log(1 - D_\delta(G_\theta(Z)))]\}, \quad (15)$$

where $D_\theta(X)$ represents the probability that X came from the data rather than P_θ . The GAN mechanism trains D_θ to maximize the probability of assigning the correct label to both training examples and fake samples. Simultaneously, it trains G_θ to minimize $\log(1 - D(G(z)))$. In other words, we would like to train G_θ such that P_θ is very close (in some distance/divergence) to P_r .

Despite its popularity, GAN has some inference issues. For instance, during the training, the generator may collapse to a setting where it always produces the same samples and face the fast vanishing gradient problem. Thus, training GANs is a delicate and unstable numerical and inferential task; see (Arjovsky et al., 2017). To overcome these problems, Arjovsky et al. propose the so-called Wasserstein GAN (WGAN) based on Wasserstein distance of order 1. The main idea is still based on mutual learning, but rather than using a discriminator to predict the probability of generated images as being real or fake, the WGAN replaces D_θ with a function f_ξ (it corresponds to ψ in Kantorovich-Rubenstein duality), indexed by parameter ξ , which is called "a critic", namely a function that evaluates the realness or fakeness of a given sample. In mathematical form, the WGAN objective function is

$$\min_{\theta} \sup_{\|f_\xi\|_L \leq 1} \{E[f_\xi(X)] - E[f_\xi(G_\theta(Z))]\}, \quad (16)$$

Also in this new formulation, the task is still to train G_θ in such a way that P_θ is very close (now, in Wasserstein distance) to P_r . (Arjovsky et al., 2017) explain how the WGAN is connected to minimum distance estimation. We remark that (16) has the same form as the Kantorovich-Rubenstein duality: we apply our dual form of ROBOT in Thm. 1 to the WGAN to obtain a new objective function

$$\min_{\theta} \sup_{\|f_\xi\|_L \leq 1, \text{range}(f_\xi) \leq 2\lambda} \{E[f_\xi(X)] - E[f_\xi(G_\theta(Z))]\}. \quad (17)$$

The central idea is to train a RWGAN by minimizing the robust Wasserstein distance (actually, using the dual form) between real and generative data. To this end, we define a novel RWGAN model based on (17). The algorithm for training this RWGAN model is very similar to the one for WGAN available in (Arjovsky et al., 2017), to which we refer for the implementation. We label this robust GAN model as RWGAN-1. Besides this model, we propose another approach derived from (3), where we use a new neural network to represent the modified distribution and add a penalty term to (16) in order to control modification. Because of space constraint, the details of this procedure are provided in Algorithm 2 in Appendix D.2. We label this new RWGAN model as RWGAN-2. Different from (Balaji et al., 2020)'s robust GAN, which uses χ_2 -divergence to constrain the modification of distribution, our RWGAN-2 does this by making use of the TV. In the sequel, we will write RWGAN-B for the robust GAN of (Balaji et al., 2020), which we implement using the same set up as Balaji et al. We remark that the RWGAN-1 is less computationally complex than RWGAN-2 and RWGAN-B. Indeed, RWGAN-2 and RWGAN-B make use of some regularized terms, thus an additional neural network is needed to represent the modification of the distribution. In contrast, RWGAN-1 has a simpler structure: for its implementation, it requires only a small modification of the activation function in the generator network; see Appendix D.2.

To investigate the robustness of RWGAN-1 and RWGAN-2 we rely on synthetic data. We consider reference samples generated from a simple model, containing n points in total, with $n - n_1$ outliers, whose data generation

process is

$$\begin{aligned} X_{i_1}^{(n)} &\sim \mathcal{U}(0, 1), X_{i_2}^{(n)} = X_{i_1}^{(n)} + 1, \\ X_i^{(n)} &= (X_{i_1}^{(n)}, X_{i_2}^{(n)}), i = 1, 2, \dots, n_1, \\ X_i^{(n)} &= (X_{i_1}^{(n)}, X_{i_2}^{(n)} + \eta), i = n_1 + 1, n_1 + 2, \dots, n, \end{aligned} \quad (18)$$

with η representing the size of outliers. We set $n = 1000$ and try four different settings by changing values of $\varepsilon = (n - n_1)/n$ and η . As it is common in the GAN literature, our generative adversarial models are obtained using the `Python` library `Pytorch`; see Appendix D.2. We display the results in Figure 3, where we compare WGAN, RWGAN-1, RWGAN-2, RWGAN-B, and RWGAN-N (based on the outlier-robust Wasserstein distance $W_1^{(\varepsilon)}$ of (Nietert et al., 2022), with $\varepsilon = 0.07$ and $\varepsilon = 0.25$, as in Nietert et al. paper) and RWGAN-D (based on the BL distance, as in (Dudley, 1969)). To measure the distance between the data simulated by the generator and the input data, we report the Wasserstein distance of order 1. For visual comparison, we display the cloud of clean data points (blue triangles) and the cloud of GAN generated points (red squares). The plots reveal that WGAN is greatly affected by outliers. Differently, RWGAN-2, RWGAN-B, RWGAN-N and RWGAN-D are able to generate data roughly consistent with the uncontaminated distribution in most of the settings. Nevertheless, they still produce some obvious abnormal points, especially when the proportion and size of outliers increase. In particular, we notice that RWGAN-D performs well in the first two contamination settings ($\varepsilon = 0.1, \eta = 2$ and $\varepsilon = 0.1, \eta = 3$) but it breaks down (namely, it generates points which are very different from the uncontaminated data) in the third and fourth contamination setting ($\varepsilon = 0.2, \eta = 2$ and $\varepsilon = 0.2, \eta = 3$), where the amount and the size of contamination increase and affect negatively the GAN performance. In a very different way, RWGAN-1 performs better than its competitors and generates data that agree with the uncontaminated distribution, even when the proportion and size of outliers are large. Several repetitions of the experiment lead to the same conclusions.

To conclude this experiment, we remark that RWGAN-N is very sensitive to the specified value of ε . For instance, in the first contaminations setting, comparing the W_1 between the clean and the generated data by RWGAN-N with $\varepsilon = 0.07$ to the W_1 of the RWGAN-1 illustrates that our method (whose λ has been selected using the procedure described in Section 6) performs better. Increasing ε to 0.25 improves on the RWGAN-N performance, making its W_1 closer to the one of the RWGAN-1. This is an important methodological consideration, which has a practical implication for the application of the RWGAN-N: a selection criterion for the ε hyper-parameter needs to be proposed, discussed and studied similarly to what we do in Section 6. To the best of our knowledge, such a criterion is not available in the literature.

Fashion-MNIST real data example. We illustrate the performance of RWGAN-1 and RWGAN-2 through comparing them with the WGAN and RWGAN-B in analysing the Fashion-MNIST real dataset, which contains 70000 grayscale images of apparels, including T-shirt, jeans, pullover, skirt, coat, etc. Each image is of 28×28 pixels, and each pixel gets value from 0 to 255, indicating the darkness or lightness. We generate outlying images by taking the negative effect of images already available in the dataset—a negative image is a total inversion, in which light areas appear dark and vice versa; in the Fashion-MNIST dataset, for each pixel of a negative image, it takes 255 minus the value of the pixel corresponding to the normal picture. By varying the number of negative images we define different levels of contamination: we consider 0, 3000, and 6000 outlying images. For the sake of visualization, in Figure 4(a) and 4(b) we show normal and negative images, respectively. Our goal is to obtain a generator which can produce images of the same style as Figure 4(a), even if it is trained on sample containing normal images and outliers. A more detailed explanation on how images generation works via `Pytorch` is available in Appendix D.2.

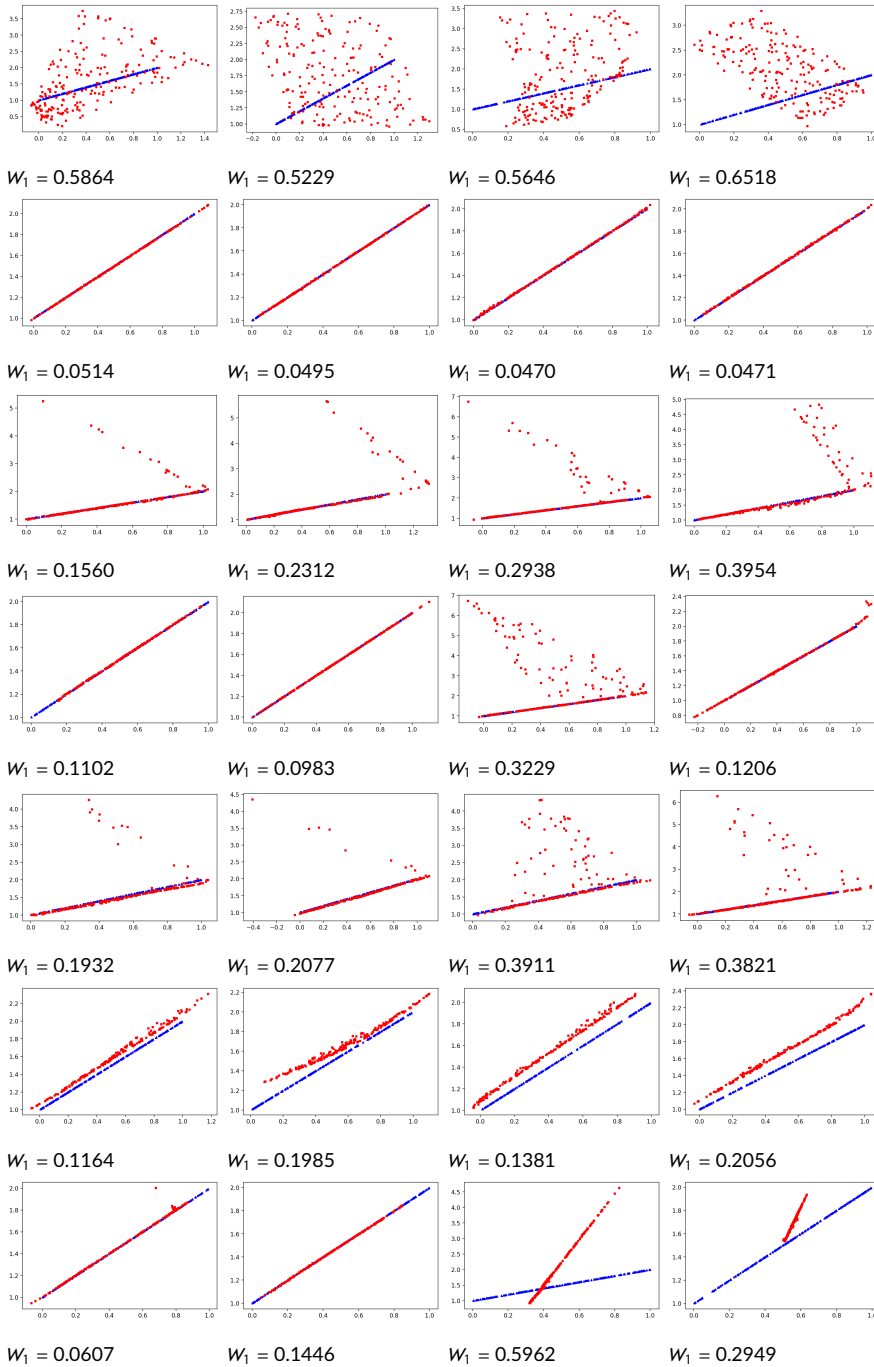


FIGURE 3 Blue triangles: 200 data points sampled from the reference distribution (uncontaminated observations). Red squares: 200 data points generated from WGAN (1st row), RWGAN-1 (2nd row), RWGAN-2 (3rd row), RWGAN-B (4th row), RWGAN-N with $\epsilon = 0.07$ (5th row), RWGAN-N with $\epsilon = 0.25$ (6th row) and RWGAN-D (7th row). Empirical Wasserstein distance $p = 1$ between triangles and squares is provided. (1st column: $\epsilon = 0.1, \eta = 2$; 2nd column: $\epsilon = 0.1, \eta = 3$; 3rd column: $\epsilon = 0.2, \eta = 2$; 4th column: $\epsilon = 0.2, \eta = 3$).

In Figure 5, we display images generated by the WGAN, RWGAN-1, RWGAN-2 and RWGAN-B under the different levels of contamination—we select λ making use of the data driven procedure described in Section 6. We see that all of the GANs generate similar images when there are no outliers in the reference sample (1st row). When there are 3000 outliers in the reference sample (2nd row), the GANs generate some negative images. However, by a simple eyeball search on the plots, we notice that WGAN produces many obvious negative images, while RWGAN-B, RWGAN-1 and RWGAN-2 generate less outlying images than WGAN. Remarkably, RWGAN-1 produces the smaller number of negative images among the considered methods. Similar conclusions holds for the higher contamination level with 6000 outlying images.

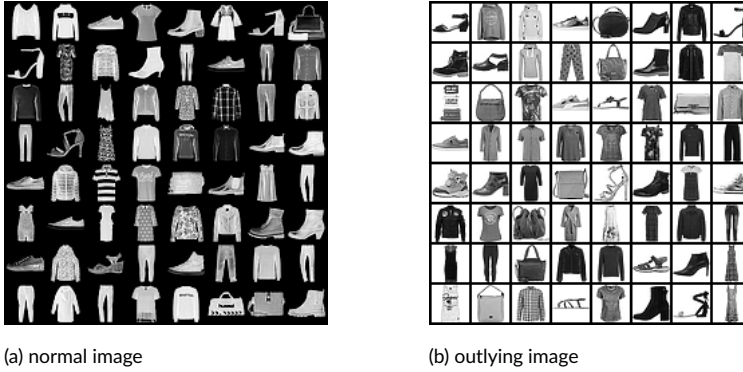


FIGURE 4 Style of clean reference sample images and outlier images. Panel (a) contains pictures that are regarded as the right style. Panel (b) contains negative pictures (outliers).

To quantify the accuracy of these GANs, we train a convolutional neural network (CNN) to classify normal images and negative images, using a training set of 60000 normal images and 60000 negative images. The resulting CNN is 100% accurate on a test set of size 20000. Then, we use this CNN to classify 1000 images generated by the four GANs. In Table 3, we report the frequencies of normal images detected by the CNN. The results are consistent with the eyeball analysis of Figure 5: the RWGAN-1 produces the smaller numbers of negative images at different contamination levels.

	WGAN	RWGAN-1	RWGAN-2	RWGAN-B
0 outliers	1.000	1.000	1.000	1.000
3000 outliers	0.941	0.982	0.953	0.945
6000 outliers	0.890	0.975	0.913	0.897

TABLE 3 Frequency of normal images output for each GAN model, under different contamination levels (the results are based on the classification by a trained CNN model).

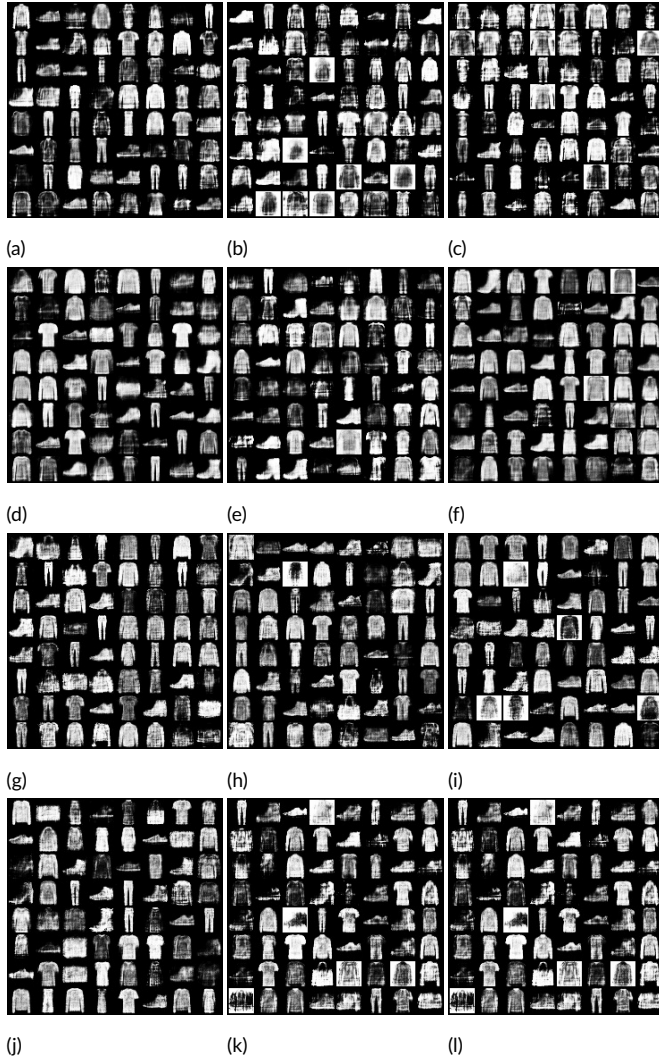


FIGURE 5 64 images generated from WGAN (1st row), RWGAN-1 (2nd row), RWGAN-2 (3rd row), RWGAN-B (4th row). (1st column: no outliers; 2nd column: 3000 outliers; 3rd column: 6000 outliers).

5.4 | Domain adaptation

Methodology. We consider an application of ROBOT to domain adaptation (DA), which is a popular machine learning problem and it deals with the inconsistent probability distributions of training samples and test samples. It has a wide range of applications in natural language processing ((Ramponi and Plank, 2020)), text analysis ((Daumé III, 2009)) and medical data analysis ((Hu et al., 2020)).

The DA problem can be stated as follows. Let

$$\{(X_i^s, Y_i^s); 1 \leq i \leq N_s\}$$

denote an i.i.d. sequence of *source domain (training sample)* and

$$\{(X_j^t, Y_j^t); 1 \leq j \leq N_t\}$$

denote an i.i.d. sequence of *target domain (test sample)*, which have joint distributions P_s and P_t , respectively. Interestingly, P_s and P_t may be two different distributions: discrepancies (also known as drift) in these data distributions depend on the type of application considered—e.g. in image analysis, domain drifts can be due to changing lighting conditions, recording device failures, or to changes in the backgrounds.

In the context of DA, we are interested in estimating the labels $Y_j^t, j = 1, \dots, N_t$, from observations

$$\{(X_i^s, Y_i^s); 1 \leq i \leq N_s\}$$

and $\{X_j^t; 1 \leq j \leq N_t\}$. Therefore, the goal is to develop a learning method which is able to transfer knowledge from the source domain to the target domain, taking into account that the training and test samples may have different distributions.

To achieve this inference goal, (Courty et al., 2017) assume that there is a transformation $\mathcal{T}$ between P_s and P_t , and estimate the map $\mathcal{T}$ via OT. To elaborate further, let us consider that, in practice, the population distribution of the source sample is unknown and it is usually estimated by its empirical counterpart

$$\hat{P}_s^{(N_s)} = N_s^{-1} \sum_{i=1}^{N_s} \delta_{X_i^s, Y_i^s}.$$

As far as the target domain is concerned, suppose we can obtain a function f to predict the unobserved response variable Y^t . Thus, the empirical distribution of the target sample is

$$\hat{P}_{t:f}^{(N_t)} = N_t^{-1} \sum_{j=1}^{N_t} \delta_{X_j^t, f(X_j^t)}.$$

The central idea of Curtey et al. (2014) relies on transforming the data to make the source and target distributions similar (namely, close in some distance), and use the label information available in the source domain to learn a classifier in the transformed domain, which in turn can be exploited to get the labels in the target domain.

More formally, the DA problem is written as an optimization problem which aims at finding the function f that minimizes the transport cost between the estimated source distribution $\hat{P}_s^{(N_s)}$ and the estimated joint target distribution

$\hat{P}_{t,f}^{(N_t)}$. To set up this problem, (Courty et al., 2014) propose to use the cost function

$$D(x_1, y_1; x_2, y_2) = \alpha d(x_1, x_2) + L(y_1, y_2),$$

which defines a joint cost combining both the distances between the features $d(x_1, x_2)$ and a measure L of the discrepancy between y_1 and y_2 , where (as recommended in the Courty et al. code) $\alpha = 0.3$. Then, the optimization problem formulated in terms of 1-Wasserstein distance is

$$\min_f W_1(\hat{P}_s^{(N_s)}, \hat{P}_{t,f}^{(N_t)}) \equiv \min_{f, \pi} \sum_{ij} D(X_i^s, Y_i^s; X_j^t, f(X_j^t)) \pi_{ij}^{(N_s, N_t)}, \quad (19)$$

which, in the case of squared loss function L (see (Courty et al., 2014)), becomes

$$\min_f N_t^{-1} \sum_{j=1}^{N_t} \|\hat{Y}_j - f(X_j^t)\|^2, \text{ where } \hat{Y}_j = N_t \sum_{i=1}^{N_s} \pi_{ij}^{(N_s, N_t)} Y_i^s,$$

Courty et al. 2014 propose to solve this problem using (a regularized version of) OT. Since OT can be sensitive to outliers, in the same spirit of (Courty et al., 2014) but with the additional aim of reducing the impact of outlying values, we propose to use ROBOT instead of OT in the DA problem. The basic intuition is that ROBOT yields a matrix $\pi^{(N_s, N_t)}$ which is able (by construction) to identify and downweight the impact of anomalous values, yielding a robust $\hat{Y}_j$, for every j . We provide the key mathematical and implementation details in Algorithm 1.

Data: source sample $\{(X_i^s, Y_i^s), 1 \leq i \leq N_s\}$, target features $\{X_j^t, 1 \leq j \leq N_t\}$, robust regularization parameter λ , parametric model f_θ (index by a parameter θ)

Result: Model f_θ

- 1 estimate θ using the source sample;
- 2 **while** θ does not convergence **do**
- 3 let $C_{ij}^{(\lambda, N_s, N_t)} = \min \left\{ D(X_i^s, Y_i^s; X_j^t, f_\theta(X_j^t)), 2\lambda \right\}$;
- 4 based on the cost matrix $(C_{ij}^{(\lambda, N_s, N_t)})_{1 \leq i \leq N_s, 1 \leq j \leq N_t}$, compute the optimal transport plan matrix $\pi^{(N_s, N_t)}$, and collect the set of all indices

$$\mathcal{I} = \{(i, j) : D(X_i^s, Y_i^s; X_j^t, f_\theta(X_j^t)) \geq 2\lambda, 1 \leq i \leq N_s, 1 \leq j \leq N_t\}$$
- 5 ;

$$s(i) = - \sum_{j=1}^{N_t} \pi_{ij}^{(N_s, N_t)} 1_{(i,j) \in \mathcal{I}}, \text{ and find } \mathcal{H}, \text{ the set of all the indices where } s(i) + 1/N_s = 0;$$
- 6 remove ℓ -th row of $\pi^{(N_s, N_t)}$ for all $\ell \in \mathcal{H}$ and normalize the matrix over columns to form a new matrix $\pi^{(N'_s, N_t)}$;
- 7 let $L_\theta = N_t^{-1} \sum_{j=1}^{N_t} \|\hat{Y}_j - f_\theta(X_j^t)\|^2$, where $\hat{Y}_j = N_t \sum_{i=1}^{N'_s} \pi_{ij}^{(N'_s, N_t)} Y_i^s$;
- 8 update θ , which minimizes L_θ ;
- 9 **end**

Algorithm 1: ROBUST DOMAIN ADAPTATION

Numerical exercise. The next Monte Carlo experiment illustrates the use and the performance of Algorithm 1

on synthetic data. We consider N_s source data points (training sample) generated from

$$\begin{aligned}
 X_i^{(N_s)} &\sim \mathcal{N}(-2, 1), i = 1, 2, \dots, [N_s/2], \\
 X_i^{(N_s)} &\sim \mathcal{N}(2, 1), i = [N_s/2] + 1, [N_s/2] + 2, \dots, N_s, \\
 Y_i^{(N_s)} &= \sin(X_i^{(N_s)}/2) + Z_i^{(N_s)}, i = 1, 2, \dots, [9N_s/10], \\
 Y_i^{(N_s)} &= \sin(X_i^{(N_s)}/2) + 2 + Z_i^{(N_s)}, i = [9n/10] + 1, [9N_s/10] + 2, \dots, N_s, \\
 Z_i^{(N_s)} &\sim \mathcal{N}(0, 0.1^2)
 \end{aligned} \tag{20}$$

and the N_t target points (test sample) are generated from

$$\begin{aligned}
 X_j^{(N_t)} &\sim \mathcal{N}(-1, 1), j = 1, 2, \dots, [N_t/2], \\
 X_j^{(N_t)} &\sim \mathcal{N}(2, 1), j = [N_t/2] + 1, [N_t/2] + 2, \dots, N_t, \\
 Y_j^{(N_t)} &= \sin((X_j^{(N_t)} - 2)/2) + Z_j^{(N_t)}, j = 1, 2, \dots, N_t, \\
 Z_j^{(N_t)} &\sim \mathcal{N}(0, 0.1^2).
 \end{aligned} \tag{21}$$

We notice that the source and target domain have a similar distribution but there is a shift, called *drift* in the DA literature, between them. In the training sample, outliers in the response variable are due to the drift added to the $Y_i^{(N_s)}$ for $i = [9n/10] + 1, [9N_s/10] + 2, \dots, N_s$. Moreover, mimicking the problem of leverage points in linear regression (see (Hampel et al., 1986, Ch. 6)), our simulation design is such that half of the $X_i^{(N_s)}$'s are generated from a $\mathcal{N}(-2, 1)$, whilst the other half is obtained from a shifted/drifted Gaussian $\mathcal{N}(2, 1)$; the feature space in the test sample contains a similar pattern but with a different drift.

In our simulation exercise, we set $N_s = 200$ and $N_t = 200$ and we compare three methods for estimating f : the first one is the Kernel Ridge Regression (KRR); the second one is the method proposed by (Courty et al., 2017), which is a combination of OT and KRR; the third one is our method, which is based on a modification of the second method, as obtained using ROBOT instead of OT. To the best of our knowledge, this application of ROBOT to DA is new to the literature.

In Figure 6 we show the results. The outliers in the training sample are clearly visible as a cluster of points which are far a part from the bulk of the data. The plot illustrates that the curve obtained via KRR fits reasonably well only the majority of the data in the source domain, but it looks inadequate for the target domain. This is due to the drift between the target and source distributions: by design, the KRR approach cannot capture this drift since it takes into account only the source domain information. The method in (Courty et al., 2017) yields a curve that is closer to the target data than the KRR curve, but because of the outliers in the $Y_i^{(N_s)}$, it does not perform well when the feature takes on negative values or values larger than four. Differently from the other methods, our method yields an estimated curve that is not too influenced by outlying values and that fits the target distribution better than the other methods.

6 | CHOICE OF λ VIA CONCENTRATION INEQUALITIES

The application of $W^{(\lambda)}$ requires the specification of λ . In the literature on robust statistics, many selection criteria are based on first-order asymptotic theory. For instance, (Mancini et al., 2005) and (La Vecchia et al., 2012) propose to select the tuning constant for Huber-type M-estimators achieving a given asymptotic efficiency at the reference model (namely, without considering the contamination). Unfortunately, these type of criteria hinge on the asymptotic notion of influence function, which is not available for our estimators defined in Section 4; see (Bassetti and Regazzini,

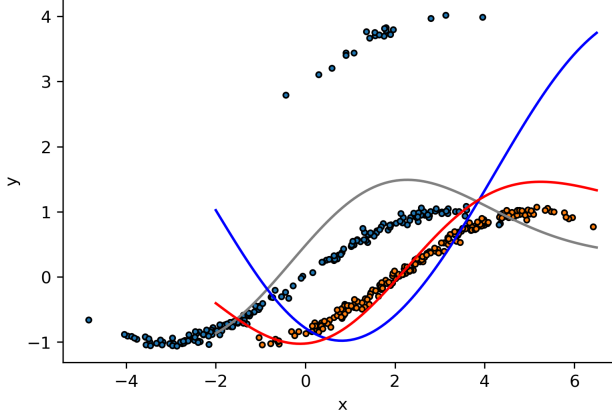


FIGURE 6 Scatter plots of the source and target samples, and functions estimated by different approaches. The grey-blue points represent the source samples, while the orange points represent the target samples. The gray, blue and red solid curves are obtained by KRR, the method of (Courty et al., 2017) and our method, respectively.

2006) for a related discussion on MKE. (Mukherjee et al., 2021) propose a criterion which needs access to a clean data set and do not develop its theoretical underpinning. To fill this gap in the literature, we introduce a selection criterion of λ that does not have an asymptotic nature and is data-driven. Moreover, it does not need the computation of $\hat{\theta}_n^\lambda$ for each λ , that makes its implementation fast.

To start with, let us notice that the term $W^{(\lambda)}(\mu_{\hat{\theta}_n^\lambda}, \mu_\star)$ expresses the robust Wasserstein distance between the fitted model ($\mu_{\hat{\theta}_n^\lambda}$) and the actual data generating measure ($\mu_\star$). Intuitively, outlying values induce bias in the estimates, entailing abrupt changes in the distribution of $\hat{\theta}_n^\lambda$. This, in turn, entails changes in $W^{(\lambda)}(\mu_{\hat{\theta}_n^\lambda}, \mu_\star)$. Therefore, the stability of the estimates and the stability of $W^{(\lambda)}(\mu_{\hat{\theta}_n^\lambda}, \mu_\star)$ are tied together.

This consideration is the stepping stone of our selection criterion for λ . To elaborate further, repeated applications of the triangle inequality yield

$$\begin{aligned} W^{(\lambda)}(\mu_{\hat{\theta}_n^\lambda}, \mu_\star) &\leq W^{(\lambda)}(\mu_{\hat{\theta}_n^\lambda}, \hat{\mu}_n) + W^{(\lambda)}(\hat{\mu}_n, \mu_\star) \leq W^{(\lambda)}(\mu_{\hat{\theta}^\star}, \mu_\star) + 2W^{(\lambda)}(\hat{\mu}_n, \mu_\star) \\ &\leq W^{(\lambda)}(\mu_{\hat{\theta}^\star}, \mu_\star) + 2E[W^{(\lambda)}(\hat{\mu}_n, \mu_\star)] + 2|W^{(\lambda)}(\hat{\mu}_n, \mu_\star) - E[W^{(\lambda)}(\hat{\mu}_n, \mu_\star)]|, \end{aligned} \quad (22)$$

which implies that the term $W^{(\lambda)}(\hat{\mu}_n, \mu_\star) - E[W^{(\lambda)}(\hat{\mu}_n, \mu_\star)]$ has an influence on the behavior of $W^{(\lambda)}(\mu_{\hat{\theta}_n^\lambda}, \mu_\star)$. Therefore, we argue that to reduce the impact of outliers in the distribution of $W^{(\lambda)}(\hat{\mu}_n, \mu_\star)$ (hence, on the distribution of $\hat{\theta}_n^\lambda$), we need to control the concentration of $W^{(\lambda)}(\hat{\mu}_n, \mu_\star)$ about its mean. To study this quantity, we make use of (9) and (10) and we define a data-driven and non-asymptotic (i.e. depending on the finite n , on the estimated σ and on the guessed $|O|$) selection criterion for λ .

We refer to Appendix C.2 for the mathematical detail, here, we simply state the basic intuition. Our idea hinges on the fact that λ controls the impact that the outliers have on the concentration bounds of $W^{(\lambda)}(\hat{\mu}_n, \mu_\star)$. Thus, one may think of selecting the value(s) of λ ensuring a desired level of stability in the distribution of $W^{(\lambda)}(\hat{\mu}_n, \mu_\star)$. Operationally, we encode this stability in the ratio between the quantiles of $W^{(\lambda)}(\hat{\mu}_n, \mu_\star)$ in the presence and in the absence of contamination (see the quantity Q , as available in Appendix C.2). Then, we measure the impact that different values of λ have on this ratio computing its derivative w.r.t. λ (see the quantity $\partial Q / \partial \lambda$, as available in Appendix C.2). Finally, we specify a tolerance for the changes in this derivative and we identify the value(s) of λ yielding the desired level of stability.

The described procedure allows to identify a (range of values of) λ , not too close to zero and not too large either. This is a suitable and sensible achievement. Indeed, on the one hand, values of λ too close to zero imply that $W^{(\lambda)}$ trims too many observations which likely include inliers and most outliers. On the other hand, too large values of λ imply that $W^{(\lambda)}$ trims a very limited number of outliers; see Appendix C.1 for a numerical illustration of this aspect in terms of MSE as a function of λ . Additional methodological aspects of the selection procedure and its numerical illustration are deferred to Appendix C.1 and Appendix C.2.

7 | CONCLUSION

This paper examines the key theoretical aspects of ROBOT and introduces a novel class of related estimators, whose statistical properties are derived. Our numerical exercises demonstrate the advantages of our proposed inference approach. We anticipate that our findings can be extended in many other research directions. For example, one could investigate the robustness of the MRWE in an asymptotic framework, e.g. in terms of Huber's maxbias or minimax theory; see Huber and Ronchetti, 2011, Chapters 3 and 4. Additionally, it might be valuable to study the convergence of the estimator as $\epsilon \rightarrow 0$ with $n \rightarrow \infty$, (shrinking neighborhood). These investigations would require the introduction of Influence Function (IF), a mathematical tool typically applied to assess the robustness of M-estimators. The same tool could be of help to derive the asymptotic distribution of the MRWE; see e.g. (Hampel et al., 1986) and (Huber and Ronchetti, 2011). The derivation of the IF for the MRWE (and in general for MKE) is a challenging task, as highlighted by (Bassetti and Regazzini, 2006). In the near future, we plan to work on it and address some of the mentioned research challenges.

Overall, we hope our results inspire interest in combining ideas from robust statistics with other mathematics-based research topics (in particular OT), that are routinely applied in machine learning and computer science literature.

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